

Multiple Choice

1. **Answer:** A

Options: A) If the patient chooses it.; B) If the patient is bedridden.; C) If urine does not need to be measured.; D) If the patient has bladder sensation.; E) If the patient suffers from urinary incontinence.

Note: The correct answer is based on the principle of patient autonomy, which emphasizes that medical decisions, including the use of a catheter valve, should align with the patient's preferences and choices.

2. **Answer:** A

Options: A) loss of taste over the anterior two-thirds of the tongue.; B) complete loss of taste sensation.; C) paralysis of the muscles of the throat.; D) paralysis of the muscles of the jaw.; E) loss of somaesthetic sensation over the posterior two thirds of the tongue.

Note: The glossopharyngeal nerve is responsible for sensory innervation, including taste sensation from the posterior one-third of the tongue, so damage to this nerve would result in loss of taste sensation in that region, not the anterior two-thirds, which is innervated by the facial nerve.

3. **Answer:** A

Options: A) Zellweger syndrome; B) Maple syrup urine disease; C) Hemophilia; D) Medium chain acyl-CoA dehydrogenase deficiency; E) Huntington's disease

Note: Zellweger syndrome is a peroxisomal disorder characterized by the absence or dysfunction of peroxisomes, leading to impaired lipid metabolism and the accumulation of toxic substances in the body.

4. **Answer:** A

Options: A) motor and autonomic neuronal processes.; B) both autonomic and motor neuronal processes.; C) both sensory and motor neuronal processes.; D) sensory and autonomic neuronal processes.; E) motor neuronal processes.

Note: The dorsal roots of all spinal nerves contain sensory neuronal processes, while motor and autonomic neuronal processes are found in the ventral roots, making the correct answer "motor and autonomic neuronal processes" inaccurate for this question.

5. **Answer:** E

Options: A) diaphragm.; B) diaphragm and pericardium.; C) diaphragm, parietal pleura, pericardium and abdominal muscles.; D) diaphragm, parietal pleura, pericardium and lungs.; E) diaphragm, parietal pleura and pericardium.

Note: The phrenic nerves are primarily responsible for the motor innervation of the diaphragm, while also providing sensory innervation to the parietal pleura and pericardium, making the answer correct.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: Projective tests are considered valuable in assessing personality because they can reveal unconscious thoughts and feelings that may not be captured by more structured objective tests, thus providing a deeper understanding of an individual's psyche.

7. **Answer:** True

Options: A) True; B) False

Note: The removal of adhesive strips can be safely performed by the patient when they are properly instructed on the technique and conditions, ensuring minimal discomfort and preventing skin damage.

8. **Answer:** True

Options: A) True; B) False

Note: The Supreme Court has established that in age discrimination cases under the Age Discrimination in Employment Act (ADEA), plaintiffs do not need to prove that age was the sole reason for the employer's decision but only that it was a motivating factor, thus easing the burden of proof for older workers.

9. **Answer:** False

Options: A) True; B) False

Note: Recent events can have profound significance and memorability due to their immediate impact on public health, emerging research, and the evolving understanding of health issues, which shape contemporary discourse and policy.

10. **Answer:** False

Options: A) True; B) False

Note: While solving puzzles and engaging in brain-stimulating activities can enhance cognitive function in older adults, memory improvement can also be achieved through a variety of other methods, including physical exercise, proper nutrition, social engagement, and adequate sleep.

Long Form Response

11. **Answer:** Autozygosity mapping is a crucial technique in identifying disorders linked to autosomal recessive inheritance, as it leverages the presence of homozygous regions in the genome that arise from a common ancestor. This methodology involves analyzing single nucleotide polymorphisms (SNPs) across the genome to pinpoint regions of autozygosity, which indicate inherited alleles from both parents that are identical by descent. By focusing on these regions, researchers can effectively narrow down candidate genes that may harbor pathogenic mutations responsible for specific recessive disorders. The implications for health sciences are profound, as this approach not only aids in the diagnosis of genetic conditions but also enhances our understanding of population genetics, facilitates genetic counseling, and informs targeted therapies for

affected individuals, thereby improving patient outcomes.

Note: correct because it accurately describes how autozygosity mapping utilizes homozygous genomic regions inherited from a common ancestor to identify candidate genes for autosomal recessive disorders, thereby enhancing understanding and diagnosis in health sciences.

12. **Answer:** Population carrier screening for genetic disorders, while beneficial in identifying at-risk individuals, faces significant implications and limitations, particularly in the context of thalassemia. One major limitation is the high carrier frequency of thalassemia in certain populations, which can lead to a large proportion of the population being identified as carriers, complicating the interpretation of results and potential counseling. Additionally, the phenotypic variability and clinical spectrum of thalassemia can result in uncertainty regarding the severity of the disorder in carriers or their offspring, thereby challenging the utility of screening programs. Furthermore, the social and ethical considerations, including potential stigmatization and psychological impact on individuals identified as carriers, raise concerns about the implementation of such initiatives. These factors suggest that while population carrier screening can provide valuable information, the complexities surrounding thalassemia make it a less suitable candidate for broad screening initiatives compared to other genetic disorders.
- Note:* The answer correctly identifies that the high carrier frequency of thalassemia can complicate result interpretation and counseling, along with emphasizing the disorder's phenotypic variability, highlighting key limitations of population carrier screening for this genetic condition.